



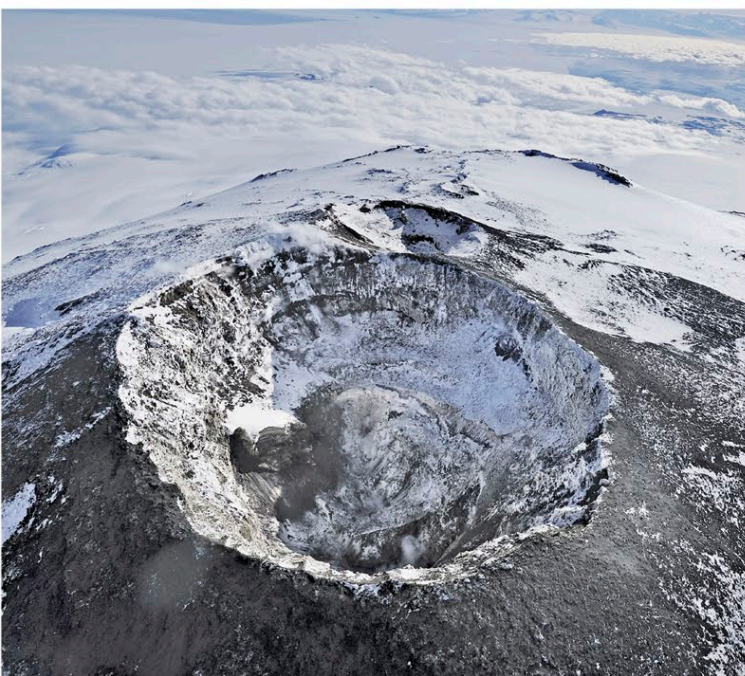
Mount Erebus

Earth's most southerly volcano known to have erupted, with a lava lake and ice caves

Antarctica's most active volcano, Mount Erebus is a 12,448-ft- (3,794-m-) high stratovolcano, located on Ross Island, just off the coast of East Antarctica. Ross Island also contains three other, apparently inactive, volcanoes.

Fire and ice

Erebus was erupting when it was first sighted by British explorer Captain James Ross in 1841, and it is still erupting frequently, if not particularly explosively, today. The volcano is one of just a handful on Earth to contain a long-lasting lava lake. This occupies a vent in the main summit crater and produces regular small explosions of lava. The temperature of the swirling pool of fiery magma is 1,700°F (900°C), and it may be several hundred feet deep. Riddling the sides of the volcano are several ice caves, carved out of its thick blanket of ice and snow by steam and other warm gases escaping from fumaroles (gas vents) on its rocky surface. Above some of the caves are ice chimneys, formed by escaping steam turning to trickles of liquid water, and then ice, on encountering the cold air outside.



△ HOUSE OF ICE

This ice cave on Mount Erebus is about 39 ft (12 m) high. It is continuously being reshaped as warm gases melt the ice in some parts, while condensing steam deposits new ice in other parts.

◁ GAPING HOLE

Erebus' summit crater is 1,300 ft (400 m) wide and 400 ft (120 m) deep. At its base are several pitlike openings, one of which contains the volcano's lava lake.

UNDERNEATH MOUNT EREBUS

Mount Erebus, as well as Mount Terror and two other currently inactive volcanoes on Ross Island, are thought to sit over plumes of hot material rising up and forming magma under the Antarctic Plate. These plumes account for the past and present volcanism.

